
PADDLEOCR-VL-CIRCUIT: BUILT FOR SCHEMATIC DIAGRAM UNDERSTANDING

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ABSTRACT

Automatic recognition of circuit schematic diagrams is a core challenge in Electronic Design Automation (EDA). This technical report presents CircuitOCR, a system based on PaddleOCR-VL-0.9B for circuit schematic OCR and netlist extraction. We construct the V5 Golden dataset of 1,857 schematics (500 synthetic + 1,357 real KiCad projects) with 100% visual-literal alignment. Through systematic LoRA ablation experiments, we identify the root cause of modality collapse—fine-tuning the vision-language projector destroys pre-trained alignment—and introduce the LLM-Only LoRA strategy that resolves it. The V10-Fixed model uses wide LoRA ($r = 16$, $\alpha = 32$, 5.7M params), fixes three critical training bugs (causal token double-shift, BPE boundary merging, `set_state_dict` silent failure in Paddle 3.1.0), and trains for 3 epochs (1,165 steps) in 43 minutes on an RTX 4060 (8 GB). Phase 1 multi-metric evaluation on easy50-pure (44 samples) yields Component F1 = 0.2061 ($4.5\times$ over baseline 0.0455), Token Recall = 0.1540 ($96\times$ over baseline 0.0016), and NED = 0.8031 (baseline 0.9296, 13.6% relative error reduction). S600 is the optimal checkpoint; S800 overfits (RepRate 40.9%). All code, weights, datasets, and a live demo are open-sourced.

Keywords Circuit Schematic · OCR · PaddleOCR-VL · LoRA Fine-tuning · Netlist Extraction · Modality Collapse

1 Introduction

Vision-Language Models (VLMs) can directly output structured text from images end-to-end, but their application to circuit schematic diagrams remains unexplored. No dedicated public benchmark exists for circuit schematic OCR. Existing datasets like Open Schematics [bshada, 2025] lack standardized evaluation protocols, while textbook-derived datasets suffer from SPICE-induced visual-to-logic annotation discrepancies.

The industrial need is substantial. The global semiconductor market exceeded \$600 billion in 2025 (SEMI/WSTS). PCB reverse engineering outsourcing alone exceeds \$500 million annually, with approximately 30% of time spent on schematic reconstruction. Use cases include hardware reverse engineering, legacy document digitization (1980s–2000s paper schematics), cross-tool migration, and the 500,000+ KiCad/EAGLE projects on GitHub lacking structured netlist exports.

Circuit schematic understanding involves two heterogeneous sub-tasks: (1) visual text recognition—extracting component designators (R1, C1), values (10k, 100nF), and net labels (VCC, GND); (2) topology graph tracing—inferring electrical connectivity from wires and net labels. Our evaluation protocol measures both dimensions: text NED for explicit recognition and component F1 for implicit structural understanding.

2 System Setup

We adopt PaddleOCR-VL-0.9B [Cui et al., 2025] (908M params), which integrates a NaViT-style dynamic vision encoder with the ERNIE-4.5-0.3B language model, supporting 109 languages. The unfine-tuned base model fails

systematically on circuit schematics: it outputs generic document text (e.g., parameter tables), memorized hallucinations, or repetitive token sequences. These failures stem from pre-training data dominated by general documents with negligible exposure to engineering drawings.

3 V5 Golden Dataset

3.1 Legacy Issues and Design Principles

Earlier dataset versions (V1–V4) had three fatal flaws: (1) GT-image misalignment—labels from KiCad S-expression parsing included invisible pin numbers and hidden net names; (2) format mismatch—Open Schematics subsets used horizontal space-concatenated format (62.1% single-word lines vs. test set’s 98.6%); (3) duplication noise—22,340 intra-line duplicate word pairs contaminating 24.3% of samples.

V5 Golden was rebuilt on two principles: (1) GT must align 100% with visible on-image text, using `draw.text()` synchronous recording for synthetic data and SVG stroked-text extraction for real KiCad data, with zero AI-generated labels; (2) format must match the test set, replacing mismatched subsets with train-real data (97.9% single-word lines), reducing duplication noise by 99.5% (22,340 $\rightarrow$ 107).

3.2 Data Sources and Statistics

Synthetic V3 (500 images). Random circuit topologies (5–30 components, 10 types) rendered as PNG, with labels from `draw.text()` logs [Zhang and Chen, 2026].

train-real (1,357 images). Real open-source KiCad projects from GitHub, rendered via `kicad-cli SVG  $\rightarrow$  cairosvg` PNG, with GT from SVG stroked-text visible elements in one-word-per-line vertical format.

Table 1: Dataset Composition and Split

Split	Synthetic V3	train-real	Total	avg Lines
Training	450	1,104	1,554	32.1
Validation	50	171	171	33.4
Test (easy50)	—	—	44	—
Test (easy100)	—	—	89	—
Total	500	1,357	1,857	—

During training, we apply online augmentations: random JPEG compression (quality 85–100), random scaling (max_dim 168–384px), and random horizontal flipping.

4 Model Fine-Tuning

4.1 Modality Collapse: Discovery and Root Cause

Initial full LoRA experiments ($r = 8$, q/k/v/o self-attention, 2.75M params on 2,433 samples) reduced NED from 0.8848 to 0.8554, but output diversity collapsed to 4%—nearly all test samples produced identical output. We define this as **Modality Collapse**.

Through six controlled experiments (E1–E6) summarized in Table 2, we systematically identified the root cause:

Core finding: Projector LoRA is the root cause. The vision-language projector (`m1p_AR`, 25.9M params) is the sole bridge from visual features to language space. LoRA perturbation distorts these features into out-of-distribution vectors the LLM cannot interpret, causing it to degenerate into repeating high-frequency training tokens. Freezing the Projector while teaching only the LLM to reinterpret existing visual features is the safe strategy.

4.2 V5: LLM-Only LoRA Validation

V5 freezes the Projector and fine-tunes only LLM self-attention ($r = 8$, $\alpha = 16$, 1.25M params), with resolution increased to 384px and learning rate 2×10^{-5} . Training on 1,857 samples for 1 epoch (928 steps, 22 min) achieved 90% output diversity—the first fine-tuned model to produce different outputs for different images. However, $r = 8$ capacity was insufficient: NED (0.9031) remained below baseline due to premature convergence at ~ 200 steps.

Table 2: Ablation Experiments for Modality Collapse

Exp	Variable	Result	Conclusion
E1	Real vs. blank image	Outputs differ	Vision encoder works
E2	Resolution 168→336px	Still collapses	Resolution not the cause
E3	1 vs. 3 epochs	NED improves 0.9%	Multi-epoch helps but insufficient
E4	Add Projector LoRA	NED drops to 0.8271	Projector is the bottleneck
E5	Rank $r = 8 \rightarrow r = 16$	NED drops to 0.7961	Capacity helps, diversity still 4%
E6	Freeze Projector	Diversity 90%	Collapse fully resolved

4.3 V10-Fixed: Three Technical Fixes

V10-Fixed scales LoRA to $r = 16$, $\alpha = 32$ (5.7M params, 0.63% of 908M), targeting 310 projection matrices across LLM self-attention and linear layers. Three critical training bugs were fixed:

1. **Causal token double-shift:** PaddleOCR-VL’s `AutoModelForConditionalGeneration` internally applies causal shift. Our manual shift caused double-shifting—supervising step t with label $t + 2$, corrupting gradients. Fixed by removing the manual shift.
2. **BPE boundary merging:** Concatenating prompt and label strings before tokenization causes BPE to merge characters across the boundary (e.g., “\n” + “R” → merged token). Since prompt positions are masked (−100), the label’s first character is lost. Fixed by tokenizing prompt and label independently and concatenating token IDs.
3. **set_state_dict silent failure:** In Paddle 3.1.0, `LoRAModel.set_state_dict()` returns `None` instead of a tuple, causing silent LoRA loading failure. Fixed by using the `LoRAModel` wrapper with manual `p.set_value()` per parameter.

Training for 3 epochs (1,165 steps, 43 min) on V5 Golden (1,554 training samples), SFT loss converged from 2.71 to 0.30.

4.4 Phase 1 Multi-Metric Evaluation

Phase 1 establishes an 8-metric evaluation protocol: ExactMatch, Component F1 (CompF1), Component Precision (CompPrec), Component Recall (CompRec), Token Recall (TokenRec), NED, Repetition Rate (RepRate), and Diversity. We evaluated all four checkpoints (S400, S600, S800, final) on easy50-pure (44 samples).

Table 3: Phase 1 Multi-Metric Benchmark (easy50-pure, 44 samples)

Model	ExactMatch	CompF1	CompPrec	CompRec	TokenRec	NED ↓	RepRate	Diversity
Base	0%	0.0455	0.0455	0.0455	0.0016	0.9296	6.8%	90.9%
S400	0%	0.1820	0.1862	0.2501	0.1302	0.8298	20.5%	95.5%
S600	0%	0.2061	0.2024	0.3114	0.1540	0.8031	15.9%	90.9%
S800	0%	0.2080	0.2862	0.1996	0.1191	0.8063	40.9%	93.2%

S600 is optimal. S800 overfits: RepRate surges to 40.9% while TokenRec drops to 0.1191.

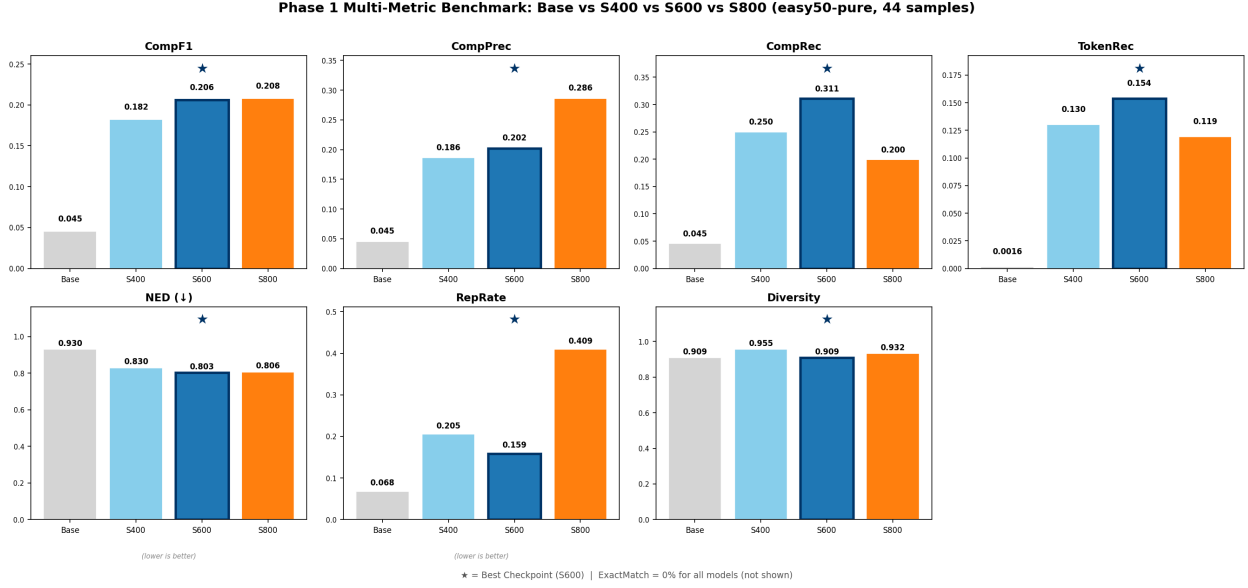


Figure 2: Phase 1 benchmark visualization across seven metrics (ExactMatch omitted: 0% for all). S600 (blue, starred) achieves the best balance. S800 (orange) shows clear overfitting.

Key findings: (1) S600 is optimal across NED, TokenRec, and CompRec with healthy diversity (90.9%). (2) S800 overfits: RepRate 40.9% (nearly $3\times$ S600), CompRec collapses to 0.1996. (3) CompF1 improves $4.5\times$ ($0.0455 \rightarrow 0.2061$), TokenRec $96\times$ ($0.0016 \rightarrow 0.1540$). (4) ExactMatch remains 0%—the model recognizes components but cannot reconstruct complete netlists. (5) No modality collapse: all checkpoints exceed 90% diversity.

Table 4: Representative model outputs on test samples (easy50-pure).

Test Image	Ground Truth (excerpt)	Base Model Output	S600 Output
benjiaomodular_sot2Hlp	/ Parameter Value Conn_01x06_Male / Parameter_a 1.0 / J3 / / Parameter_b 1.0 Conn_01x03_Male (hallucinated table) / J1 / Conn_01x03_Male		GND / R1 / 10k / C2 / 10nF / J1 / Conn_01x04_Pin
JOBriant30_PIC32	/ Conn_01x24_Pin / J2 / Conn_01x24_Pin / U2 / PIC32CX...	“(a)” (single-token collapse)	J1 / Conn_01x24_Pin / J2 / Conn_01x24_Pin / U2 / PIC32...
nxlabs-ch_shared-workflows	+5V / +3.3V / GND / J1 / GPIO / C1 / 100n / R1 / 3.9k	“(a)” (single-token collapse)	+5V / +3.3V / GND / J1 / GPIO / C1 / 100n / R1 / 3.9k

4.5 Version Progression Summary

Table 3 provides the only directly comparable evaluation—all four rows (Base, S400, S600, S800) use the same fixed `eval_benchmark_v3.py` script on the same 44-sample easy50-pure test set. Earlier versions were evaluated under different protocols and are not directly comparable to Phase 1 numbers:

- **V5 ($r = 8$, LLM-Only LoRA):** Achieved the critical breakthrough of 90% output diversity, proving the architecture direction. NED (0.9066 on 50-sample set with broken eval script) was below baseline due to insufficient LoRA capacity, but this version demonstrated that freezing the Projector prevents collapse.

- **V9-Pure ($r = 16$, Wide LoRA):** Scaled capacity to 5.7M params, achieving easy100 NED 0.7797 (baseline 0.9390) on 89 samples with the old eval script. On a 6-sample validation subset, it scored 0.8261 (baseline 0.9611). The `set_state_dict` bug was not yet discovered, so these numbers may underestimate true performance.

The Phase 1 evaluation (Table 3) supersedes all earlier results as the definitive benchmark.

5 Experimental Setup

5.1 Training Configuration

All experiments on a single NVIDIA RTX 4060 (8 GB VRAM). Final configuration: LoRA $r = 16$, $\alpha = 32$, 310 projection matrices, 5.7M params (0.63% of 908M). AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.95$, `weight_decay=0.1`), $lr = 2 \times 10^{-5}$ with cosine annealing. 3 epochs (1,165 steps), `batch_size=1`, `gradient_accumulation=4`. `max_dim=384px`, label truncation 200 chars. Training time: ~ 43 minutes.

5.2 Evaluation Metrics

Primary metric: Normalized Edit Distance $NED(p, g) = \text{Levenshtein}(p, g) / \max(|p|, |g|)$, with `--unordered` mode (lines sorted alphabetically). Phase 1 adds CompF1 (token-level F1 after line reordering and dedup), CompPrec, CompRec, TokenRec, ExactMatch, RepRate, and Diversity (unique output ratio). `RepRate > 30%` and `Diversity < 10%` indicate degenerative behavior.

6 Discussion

6.1 Modality Collapse Mechanism and Resolution

The Projector (`mlp_AR`, 25.9M params) is the sole visual-to-language bridge. Even small LoRA perturbations ($\sim 0.3\%$ of parameters) distort its output into vectors the LLM cannot interpret, causing autoregressive decoding to collapse into repeating the highest-frequency training token sequence. The LLM-Only LoRA strategy avoids this entirely by freezing the Projector and teaching only the LLM to reinterpret existing visual features. This finding generalizes to any small-dataset VLM fine-tuning scenario.

6.2 Data Quality Over Quantity

V4’s 5,686 noisy samples (format mismatch 62.1% vs. 98.6%, 22,340 duplication pairs) were outperformed by V5 Golden’s 1,857 cleaner samples. The Masala-CHAI textbook dataset [Bhandari et al., 2025] was discarded because its SPICE-derived annotations contained invisible logical nodes (VSS, GND) that caused hallucinations—removing this 30% subset actually improved evaluation scores. Clean small datasets consistently outperform noisy large ones.

6.3 Checkpoint Selection and Overfitting

The S600→S800 degradation (RepRate 15.9% → 40.9%, TokenRec 0.1540 → 0.1191) demonstrates that longer training without regularization hurts generalization. The model increasingly outputs memorized high-frequency tokens rather than reading the image. Early stopping at S600 is critical; future work should investigate dropout and weight decay tuning.

7 Conclusion

This report presents CircuitOCR with four contributions:

1. **V5 Golden Dataset:** 1,857 samples with 100% visual-literal alignment, 99.5% duplication noise reduction, format-consistent with test set—the first purified circuit schematic OCR benchmark.
2. **Modality Collapse Resolution:** E1–E6 experiments identify Projector LoRA as the root cause; LLM-Only LoRA restores diversity from 4% to 90%.
3. **V10-Fixed Model:** Wide LoRA ($r = 16$, 5.7M params) + three training fixes achieves CompF1 0.2061 (4.5 $\times$), TokenRec 0.1540 (96 $\times$), NED 0.8031 (13.6% error reduction) on easy50-pure. S600 is optimal; S800 overfits.
4. **Open-Source Ecosystem:** Dataset, training/evaluation scripts, LoRA weights, and Gradio demo fully open-sourced.

Limitations include RTX 4060 8GB compute constraints (no Flash Attention), ExactMatch remaining at 0% (no model reconstructs complete netlists), and pin-level topology evaluation not yet implemented. Future work: full topology metrics, Flash Attention on larger GPUs, SPICE verification as reward signal, and regularization to prevent S600→S800 overfitting.

Appendix

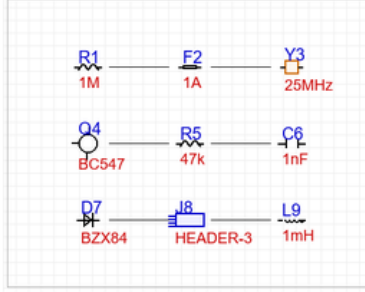
A. Raw Dataset Samples

Figure 4 shows raw (unprocessed) sample images from each data source, with ground truth labels. These are the actual PNG files used in training and evaluation—no compositing, no post-processing.

References

- bshada. Open schematics: A dataset of electronic schematics from hardware projects, 2025. URL <https://huggingface.co/datasets/bshada/open-schematics>. License: CC-BY-4.0.
- Cheng Cui, Ting Sun, Suyin Liang, Tingquan Gao, Zelun Zhang, Jiaxuan Liu, Xueqing Wang, Changda Zhou, Hongen Liu, Manhui Lin, Yue Zhang, Yubo Zhang, Handong Zheng, Jing Zhang, Jun Zhang, Yi Liu, Dianhai Yu, and Yanjun Ma. Paddleocr-vl: Boosting multilingual document parsing via a 0.9b ultra-compact vision-language model, 2025. URL <https://arxiv.org/abs/2510.14528>.
- Jianning Zhang and Yifei Chen. A synthetic dataset for circuit schematic ocr and netlist extraction, 2026. URL <https://github.com/ZhangJ83/circuit-ocr-dataset>.
- Jitendra Bhandari, Vineet Bhat, Yuheng He, Hamed Rahmani, Siddharth Garg, and Ramesh Karri. Masala-chai: A large-scale spice netlist dataset for analog circuits by harnessing ai, 2025. URL <https://arxiv.org/abs/2411.14299>.

Synthetic V3 (Training)



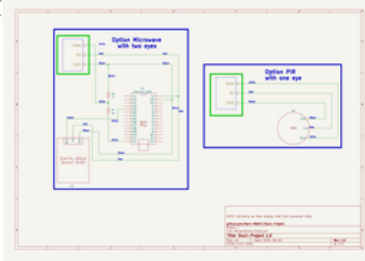
Y3
R1
F2
1M
1A
BC547
C6
R5
D7

train-real KiCad (Training)

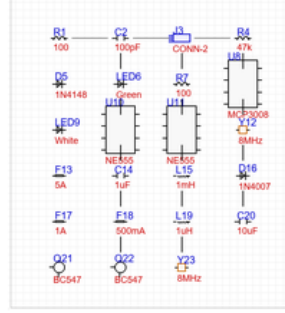


R1
100
Q1
BC547
C1
R2
47k

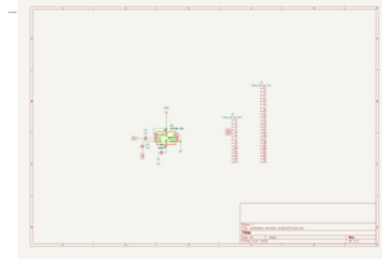
Test (easy50-pure)



U2
~
A1
Arduino_USB
R4
3K
U3
~
...



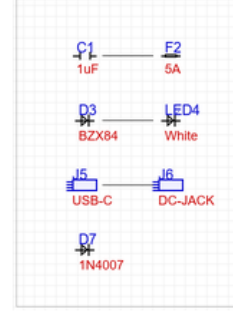
J3
R1
C2
R4
100
100F
47k
C1



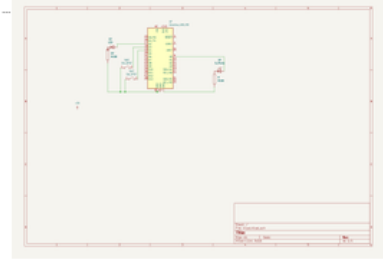
Vin
GND
C2
1nF
J2
Conn_01x14_Pin
GND
C1



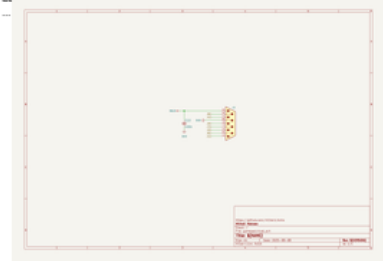
C1
10uF
C2
1uF
R1
100
F1
2A
...



C1
F2
1uF
5A
D3
LED4
BZX84
White



D7
D_Photo
SW7
SW_SPST
SW7
SW_SPST
D7
LED7



VBUS
J3
D-SUB 9 Male
GND
C10
100n
GND

Figure 1: Dataset overview: three data sources. **Top row**—Synthetic V3: programmatically generated with randomized topologies (5–30 components, 10 types), GT recorded synchronously via `draw.text()`. **Middle row**—train-real (KiCad): real open-source projects from GitHub, rendered via `kiCad-cli SVG→cairosvg PNG`, GT extracted from SVG stroked-text. **Bottom row**—Test set (easy50-pure): held-out real schematics for evaluation. All images are raw PNG renders with zero AI-generated labels.

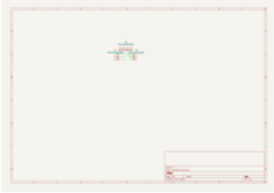
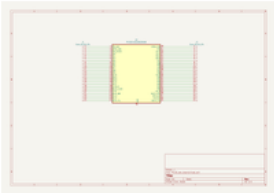
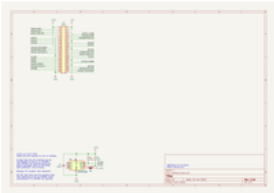
Original Image	Ground Truth	Base Model Output	S600 Output
	J2 Conn_01x06_Male J3 Conn_01x03_Male J1 Conn_01x03_Male	Parameter Value Parameter_a 1.0 Parameter_b 1.0 Parameter_c 1.0	J1 Conn_01x05_Male J2 Conn_01x05_Male J3 Conn_01x05_Male
		NED=0.8732	NED=0.1071
	J1 Conn_01x24_Pin J2 Conn_01x24_Pin U2 PIC32CX1012BZ25048	-	J1 Conn_01x_Odd_E_even J2 Conn_01x_Odd_E_even
		NED=1.0000	NED=0.6318
	+5V +3.3V GND GND J1 GPIO C1 100n R1 3.9k JP1 ID_VIP R2 3.9k +3.3V	-	GND GND GND +3.3V +3.3V +3.3V +3.3V +3.3V
		NED=1.0000	NED=0.6818

Figure 3: Model comparison: three test examples showing the original schematic, ground truth netlist, base model output (pre-fine-tuning), and S600 output (post-fine-tuning). The base model either hallucinates generic text or collapses into repetition ($NED \geq 0.87$). S600 correctly identifies component designators and values, though full netlist reconstruction remains imperfect. NED values shown at bottom-right of each output column.

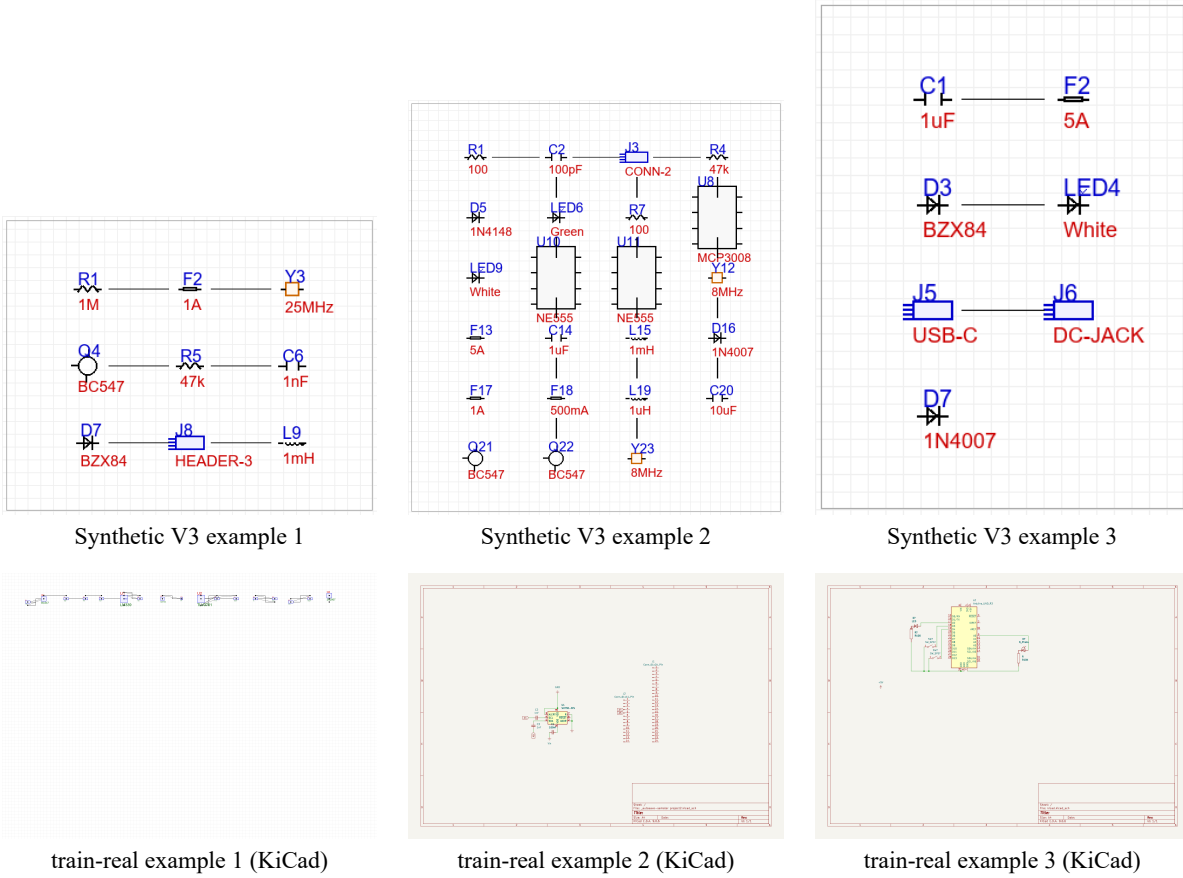


Figure 4: Raw dataset samples. Top row: Synthetic V3 (programmatic generation, GT from `draw.text()` logs). Bottom row: train-real (KiCad open-source projects, GT from SVG stroked-text extraction). All images are raw PNG renders.